

1(a)(i)	work (done) per unit mass	B1
	work (done on mass) in moving mass from infinity (to the point)	B1
1(a)(ii)	$E_P = \phi m$ $E_P = (-GM/r) \times m = -GMm/r$ or $\phi = -GM/r$ and $E_P = \phi m = -GMm/r$	B1
1(b)(i)	$\Delta E_P = 6.67 \times 10^{-11} \times 1.99 \times 10^{30} \times 2.20 \times 10^{14} \times [1/(6.38 \times 10^{10}) - 1/(8.44 \times 10^{11})]$	C1
	$= 4.23 \times 10^{23} \text{ J}$	A1
1(b)(ii)	(gravitational) force is attractive so decrease or (gravitational) force does work so decrease	B1
1(b)(iii)	$\Delta E_P = \frac{1}{2}m(v_2^2 - v_1^2)$	C1
	$4.23 \times 10^{23} = \frac{1}{2} \times 2.20 \times 10^{14} \times (v^2 - 34\,100^2)$	C1
	$v (= 70\,800 \text{ m s}^{-1}) = 70.8 \text{ km s}^{-1}$	A1
1(c)	both PE and KE equations include m , so path is unchanged	B1